

Notation and Assumptions

Suppose $\mathcal{D} = \{\mathbf{z}_i\}_{i=1:n}$ are independent and identically distributed observations. Consider a model for $\mathcal{D}$ parameterized by $\boldsymbol{\theta}$. Let $\ell(\boldsymbol{\theta})$ denote the log likelihood:

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^n \ell_i(\boldsymbol{\theta}) = \sum_{i=1}^n \ell(\boldsymbol{\theta}; \mathbf{z}_i).$$

Denote the true value of $\boldsymbol{\theta}$ by $\boldsymbol{\theta}_0$. The score for $\boldsymbol{\theta}$ is:

$$\dot{\ell}_{\boldsymbol{\theta}}(\boldsymbol{\theta}_0) \equiv \left. \frac{\partial \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0}.$$

The Hessian for $\boldsymbol{\theta}$ is:

$$\ddot{\ell}_{\boldsymbol{\theta}\boldsymbol{\theta}'}(\boldsymbol{\theta}_0) \equiv \left. \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \right|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0}.$$

Partition $\boldsymbol{\theta} = (\boldsymbol{\beta}, \boldsymbol{\alpha})$, where $\boldsymbol{\beta}$ is the target parameter, and $\boldsymbol{\alpha}$ is a nuisance parameter. The null hypothesis will take the form $H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0$. Let $\hat{\boldsymbol{\theta}} = (\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\alpha}})$ denote the unrestricted MLE of $\boldsymbol{\theta}$, which satisfies:

$$\dot{\ell}_{\boldsymbol{\beta}}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\alpha}}) = \mathbf{0}, \quad \dot{\ell}_{\boldsymbol{\alpha}}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\alpha}}) = \mathbf{0}.$$

Let $\tilde{\boldsymbol{\theta}} = (\boldsymbol{\beta}_0, \tilde{\boldsymbol{\alpha}})$ denote the restricted MLE of $\boldsymbol{\theta}$, which satisfies:

$$\dot{\ell}_{\boldsymbol{\alpha}}(\boldsymbol{\beta}_0, \tilde{\boldsymbol{\alpha}}) = \mathbf{0}.$$

Assume sufficient regularity that, under H_0 , the restricted and unrestricted MLEs are each $\sqrt{n}$ -consistent:

$$\tilde{\boldsymbol{\theta}} = \boldsymbol{\theta}_0 + \mathcal{O}_p(n^{-1/2}), \quad \hat{\boldsymbol{\theta}} = \boldsymbol{\theta}_0 + \mathcal{O}_p(n^{-1/2}).$$

The score equation converges in distribution as:

$$n^{-1/2} \dot{\ell}_{\boldsymbol{\theta}}(\boldsymbol{\theta}_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \dot{\ell}_{\boldsymbol{\theta},i}(\boldsymbol{\theta}_0) \xrightarrow{\mathcal{L}} N\{\mathbf{0}, \mathbf{B}(\boldsymbol{\theta}_0)\},$$

where $\mathbf{B}$ is the expected outer product of the score:

$$\mathbf{B}_0 \equiv \mathbf{B}(\boldsymbol{\theta}_0) \equiv \mathbb{E}_Z\{\dot{\ell}_{\boldsymbol{\theta},i}(\boldsymbol{\theta}_0) \otimes \dot{\ell}_{\boldsymbol{\theta},i}(\boldsymbol{\theta}_0)\}.$$

The negative Hessian converges in probability as:

$$-n^{-1} \ddot{\ell}_{\boldsymbol{\theta}\boldsymbol{\theta}'}(\boldsymbol{\theta}_0) = -\frac{1}{n} \sum_{i=1}^n \ddot{\ell}_{\boldsymbol{\theta}\boldsymbol{\theta}',i}(\boldsymbol{\theta}_0) = \mathbf{A}(\boldsymbol{\theta}_0) + o_p(1),$$

where $\mathbf{A}$ is the expected information matrix:

$$\mathbf{A}_0 \equiv \mathbf{A}(\boldsymbol{\theta}_0) = \mathbb{E}_Z \{ -\ddot{\ell}_{\theta\theta',i}(\boldsymbol{\theta}_0) \}.$$

Moreover, suppose sufficient regularity that:

$$-n^{-1}\ddot{\ell}_{\theta\theta'}(\tilde{\boldsymbol{\theta}}) = \mathbf{A}(\boldsymbol{\theta}_0) + o_p(1), \quad -n^{-1}\ddot{\ell}_{\theta\theta'}(\hat{\boldsymbol{\theta}}) = \mathbf{A}(\boldsymbol{\theta}_0) + o_p(1).$$

Let $\boldsymbol{\Omega}_0$ denote the covariance matrix:

$$\boldsymbol{\Omega}_0 \equiv \boldsymbol{\Omega}(\boldsymbol{\theta}_0) = \mathbf{A}(\boldsymbol{\theta}_0)^{-1} \mathbf{B}(\boldsymbol{\theta}_0) \mathbf{A}(\boldsymbol{\theta}_0)^{-T}.$$

Within the exponential family $\mathbf{A}_0 = \mathbf{B}_0$ such that:

$$\boldsymbol{\Omega}_0 = \mathbf{A}_0^{-1},$$

which is the inverse expected information.

Wald Test

Proposition 2.1. Under $H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$, the Wald statistic converges as:

$$n(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)\boldsymbol{\Omega}_0^{-1}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{\mathcal{L}} \chi_r^2,$$

where $r = \dim(\boldsymbol{\theta}_0)$. ◆

Proof. Taylor expansion of the score for $\boldsymbol{\theta}$ about the truth:

$$\mathbf{0} = \dot{\ell}_{\boldsymbol{\theta}}(\hat{\boldsymbol{\theta}}) = \dot{\ell}_{\boldsymbol{\theta}}(\boldsymbol{\theta}_0) + \ddot{\ell}_{\boldsymbol{\theta}\boldsymbol{\theta}'}(\boldsymbol{\theta}_0)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + \mathcal{O}_p(\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\|^2)$$

Solving for $(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)$:

$$(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = \{ -\ddot{\ell}_{\boldsymbol{\theta}\boldsymbol{\theta}'}(\boldsymbol{\theta}_0) \}^{-1} \dot{\ell}_{\boldsymbol{\theta}}(\boldsymbol{\theta}_0) + \mathcal{O}_p(n^{-1}).$$

Scaling by $\sqrt{n}$:

$$\begin{aligned} \sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) &= \{ -n^{-1}\ddot{\ell}_{\boldsymbol{\theta}\boldsymbol{\theta}'}(\boldsymbol{\theta}_0) \}^{-1} n^{-1/2} \dot{\ell}_{\boldsymbol{\theta}}(\boldsymbol{\theta}_0) + \mathcal{O}_p(n^{-1/2}) \\ &= \mathbf{A}_0^{-1} n^{-1/2} \dot{\ell}_{\boldsymbol{\theta}}(\boldsymbol{\theta}_0) + o_p(1) \\ &\xrightarrow{\mathcal{L}} N(\mathbf{0}, \mathbf{A}_0^{-1} \mathbf{B}_0 \mathbf{A}_0^{-T}). \end{aligned}$$
■

Corollary 2.1. The Wald test of $H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0$ is:

$$n(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)\boldsymbol{\Omega}_{0,\boldsymbol{\beta}\boldsymbol{\beta}'}^{-1}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{\mathcal{L}} \chi_p^2,$$

where $p = \dim(\boldsymbol{\beta}_0)$ and $\boldsymbol{\Omega}_{0,\boldsymbol{\beta}\boldsymbol{\beta}'}$ is the sub-matrix of $\boldsymbol{\Omega}_0$ corresponding to $\boldsymbol{\beta}$. ♣

Remark 2.1. Within the exponential family $\boldsymbol{\Omega}_{0,\boldsymbol{\beta}\boldsymbol{\beta}'}^{-1}$ is the efficient information for $\boldsymbol{\beta}$:

$$\boldsymbol{\Omega}_{0,\boldsymbol{\beta}\boldsymbol{\beta}'}^{-1} = \mathbf{A}_{\boldsymbol{\beta}\boldsymbol{\beta}'} - \mathbf{A}_{\boldsymbol{\beta}\boldsymbol{\alpha}'} \mathbf{A}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'}^{-1} \mathbf{A}_{\boldsymbol{\alpha}\boldsymbol{\beta}'}.$$
◆

Score Test

Proposition 3.1. Under $H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$, the score statistic converges as:

$$\frac{1}{n} \dot{\ell}_{\boldsymbol{\theta}}(\boldsymbol{\theta}_0)' \mathbf{B}_0^{-1} \dot{\ell}_{\boldsymbol{\theta}}(\boldsymbol{\theta}_0) \xrightarrow{\mathcal{L}} \chi_r^2,$$

where $r = \dim(\boldsymbol{\theta}_0)$. ◆

Proposition 3.2. The score test of $H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0$ is:

$$\frac{1}{n} \dot{\ell}_{\boldsymbol{\beta}}(\boldsymbol{\beta}_0, \tilde{\boldsymbol{\alpha}})' \{ \mathbf{C}_0 \mathbf{B}_0 \mathbf{C}_0' \}^{-1} \dot{\ell}_{\boldsymbol{\beta}}(\boldsymbol{\beta}_0, \tilde{\boldsymbol{\alpha}}) \xrightarrow{\mathcal{L}} \chi_p^2,$$

where $p = \dim(\boldsymbol{\beta}_0)$ and

$$\mathbf{C}_0 = (\mathbf{I}, -\mathbf{A}_{\boldsymbol{\beta}\boldsymbol{\alpha}'} \mathbf{A}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'}^{-1}).$$
◆

Proof. Taylor expand the scores for $\boldsymbol{\beta}$ and $\boldsymbol{\alpha}$ about the constrained MLE:

$$\begin{aligned} \dot{\ell}_{\boldsymbol{\beta}}(\boldsymbol{\beta}_0, \tilde{\boldsymbol{\alpha}}) &= \dot{\ell}_{\boldsymbol{\beta}}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) + \ddot{\ell}_{\boldsymbol{\beta}\boldsymbol{\alpha}'}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0)(\tilde{\boldsymbol{\alpha}} - \boldsymbol{\alpha}_0) + \mathcal{O}_p(n^{-1}), \\ 0 = \dot{\ell}_{\boldsymbol{\alpha}}(\boldsymbol{\beta}_0, \tilde{\boldsymbol{\alpha}}) &= \dot{\ell}_{\boldsymbol{\alpha}}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) + \ddot{\ell}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0)(\tilde{\boldsymbol{\alpha}} - \boldsymbol{\alpha}_0) + \mathcal{O}_p(n^{-1}). \end{aligned}$$

Substituting the second expansion into the first:

$$\begin{aligned} \dot{\ell}_{\boldsymbol{\beta}}(\boldsymbol{\beta}_0, \tilde{\boldsymbol{\alpha}}) &= \dot{\ell}_{\boldsymbol{\beta}}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) - \ddot{\ell}_{\boldsymbol{\beta}\boldsymbol{\alpha}'}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) \{ \ddot{\ell}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) \}^{-1} \dot{\ell}_{\boldsymbol{\alpha}}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) \\ &= \left(\mathbf{I}, -\ddot{\ell}_{\boldsymbol{\beta}\boldsymbol{\alpha}'}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) \{ \ddot{\ell}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) \}^{-1} \right) \begin{pmatrix} \dot{\ell}_{\boldsymbol{\beta}}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) \\ \dot{\ell}_{\boldsymbol{\alpha}}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) \end{pmatrix} \\ &= \left(\mathbf{I}, -n^{-1} \ddot{\ell}_{\boldsymbol{\beta}\boldsymbol{\alpha}'} \{ n^{-1} \ddot{\ell}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) \}^{-1} \right) \dot{\ell}_{\boldsymbol{\theta}}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) \\ &= (\mathbf{I}, -\mathbf{A}_{\boldsymbol{\beta}\boldsymbol{\alpha}'} \mathbf{A}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'}^{-1}) \dot{\ell}_{\boldsymbol{\theta}}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) + o_p(1). \end{aligned}$$

Let $\mathbf{C}_0 = (\mathbf{I}, -\mathbf{A}_{\boldsymbol{\beta}\boldsymbol{\alpha}'} \mathbf{A}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'}^{-1})$, then:

$$\frac{1}{\sqrt{n}} \dot{\ell}_{\boldsymbol{\beta}}(\boldsymbol{\beta}_0, \tilde{\boldsymbol{\alpha}}) = \mathbf{C}_0 \cdot \frac{1}{\sqrt{n}} \dot{\ell}_{\boldsymbol{\theta}}(\boldsymbol{\beta}_0, \boldsymbol{\alpha}_0) + o_p(1) \xrightarrow{\mathcal{L}} N(\mathbf{0}, \mathbf{C}_0 \mathbf{B}_0 \mathbf{C}_0').$$

The asymptotic covariance is:

$$\mathbf{C}_0 \mathbf{B}_0 \mathbf{C}_0' = \mathbf{B}_{\boldsymbol{\beta}\boldsymbol{\beta}'} - \mathbf{B}_{\boldsymbol{\beta}\boldsymbol{\alpha}'} \mathbf{A}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'}^{-1} \mathbf{A}_{\boldsymbol{\beta}\boldsymbol{\alpha}'}^T - \mathbf{A}_{\boldsymbol{\beta}\boldsymbol{\alpha}'} \mathbf{A}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'}^{-1} \mathbf{B}_{\boldsymbol{\alpha}\boldsymbol{\beta}'} + \mathbf{A}_{\boldsymbol{\beta}\boldsymbol{\alpha}'} \mathbf{A}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'}^{-1} \mathbf{B}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'} \mathbf{A}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'}^{-1} \mathbf{A}_{\boldsymbol{\beta}\boldsymbol{\alpha}'}^T.$$
■

Corollary 3.1. Within the exponential family:

$$\mathbf{C}_0 \mathbf{B}_0 \mathbf{C}_0' = \mathbf{A}_{\boldsymbol{\beta}\boldsymbol{\beta}'} - \mathbf{A}_{\boldsymbol{\beta}\boldsymbol{\alpha}'} \mathbf{A}_{\boldsymbol{\alpha}\boldsymbol{\alpha}'}^{-1} \mathbf{A}_{\boldsymbol{\alpha}\boldsymbol{\beta}'} = (\mathbf{A}_0^{-1})_{\boldsymbol{\beta}\boldsymbol{\beta}'},$$

which is the efficient information for $\boldsymbol{\beta}$. ♣

Likelihood Ratio Test

Remark 4.1. This section assumes $\mathbf{A}_0$ is symmetric and positive definite. ♦

Proposition 4.1. Under the $H_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$, the likelihood ratio statistic converges as:

$$2\{\ell(\hat{\boldsymbol{\theta}}) - \ell(\boldsymbol{\theta}_0)\} \xrightarrow{\mathcal{L}} \sum_{j=1}^r \lambda_j \chi_1^2,$$

where $(\lambda_1, \dots, \lambda_r)$ are the eigenvalues of $\mathbf{A}_0^{-1/2} \mathbf{B}_0 \mathbf{A}_0^{-1/2}$ and $r = \dim(\boldsymbol{\theta}_0)$. ♦

Proof. Taylor expand the log likelihood at the truth about the unconstrained MLE:

$$\ell(\boldsymbol{\theta}_0) = \ell(\hat{\boldsymbol{\theta}}) + \dot{\ell}_{\boldsymbol{\theta}}(\hat{\boldsymbol{\theta}})(\boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}}) + \frac{1}{2}(\boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}})\ddot{\ell}_{\boldsymbol{\theta}\boldsymbol{\theta}'}(\hat{\boldsymbol{\theta}})(\boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}}) + \mathcal{O}_p(n^{-3/2}).$$

Since $\dot{\ell}_{\boldsymbol{\theta}}(\hat{\boldsymbol{\theta}}) = \mathbf{0}$, upon rearranging:

$$\begin{aligned} 2\{\ell(\hat{\boldsymbol{\theta}}) - \ell(\boldsymbol{\theta}_0)\} &= (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)' \{ -\ddot{\ell}_{\boldsymbol{\theta}\boldsymbol{\theta}'}(\hat{\boldsymbol{\theta}}) \} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + \mathcal{O}_p(n^{-3/2}) \\ &= n(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)' \{ -n^{-1}\ddot{\ell}_{\boldsymbol{\theta}\boldsymbol{\theta}'}(\hat{\boldsymbol{\theta}}) \} (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + \mathcal{O}_p(n^{-3/2}) \\ &= n(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)' \mathbf{A}_0 (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + o_p(1). \end{aligned}$$

Recall from the Wald statistic that:

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{\mathcal{L}} N(\mathbf{0}, \mathbf{A}_0^{-1} \mathbf{B}_0 \mathbf{A}_0^{-T}).$$

Consequently, the quadratic form

$$\begin{aligned} 2\{\ell(\hat{\boldsymbol{\theta}}) - \ell(\boldsymbol{\theta}_0)\} &= n(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)' \mathbf{A}_0 (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + o_p(1) \\ &= \{\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)\}' \mathbf{A}_0 \{\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0)\} \xrightarrow{\mathcal{L}} \boldsymbol{\omega}' \mathbf{A}_0 \boldsymbol{\omega} \end{aligned}$$

where $\boldsymbol{\omega} \sim N(\mathbf{0}, \mathbf{A}_0^{-1} \mathbf{B}_0 \mathbf{A}_0^{-1})$. Let $\mathbf{L}\mathbf{L}'$ denote the Cholesky decomposition of $\mathbf{A}_0$, then:

$$\boldsymbol{\omega}' \mathbf{A}_0 \boldsymbol{\omega} = \boldsymbol{\omega}' \mathbf{L}\mathbf{L}' \boldsymbol{\omega} \stackrel{d}{=} \sum_{j=1}^r \lambda_j \chi_1^2,$$

where $(\lambda_1, \dots, \lambda_r)$ are the eigenvalues of $\mathbf{L}' \mathbf{A}_0^{-1} \mathbf{B}_0 \mathbf{A}_0^{-1} \mathbf{L}$. ■

Corollary 4.1. Within the exponential family:

$$\mathbf{A}_0^{-1/2} \mathbf{B}_0 \mathbf{A}_0^{-1/2} = \mathbf{A}_0^{-1/2} \mathbf{A}_0 \mathbf{A}_0^{-1/2} = \mathbf{I}.$$

Consequently, $\lambda_j = 1$ for each j , and:

$$2\{\ell(\hat{\boldsymbol{\theta}}) - \ell(\boldsymbol{\theta}_0)\} \xrightarrow{\mathcal{L}} \chi_r^2.$$



Proposition 4.2. The likelihood ratio test $H_0 : \beta = \beta_0$ is:

$$2\{\ell(\hat{\theta}) - \ell(\tilde{\theta})\} \xrightarrow{\mathcal{L}} \sum_{j=1}^p \lambda_j \chi_1^2,$$

where $(\lambda_1, \dots, \lambda_p)$ are the eigenvalues of:

$$(\mathbf{A}_0^{-1})_{\beta\beta'}^{1/2} \mathbf{C}_0 \mathbf{B}_0 \mathbf{C}_0' (\mathbf{A}_0^{-1})_{\beta\beta'}^{1/2},$$

$\mathbf{C}_0 = (\mathbf{I}, -\mathbf{A}_{\beta\alpha'} \mathbf{A}_{\alpha\alpha'}^{-1})$, and $p = \dim(\beta_0)$. ◆

Proof. Taylor expand the log likelihood at the constrained MLE about the unconstrained MLE:

$$\ell(\tilde{\theta}) = \ell(\hat{\theta}) + \dot{\ell}_{\theta}(\hat{\theta})(\tilde{\theta} - \hat{\theta}) + \frac{1}{2}(\tilde{\theta} - \hat{\theta})' \ddot{\ell}_{\theta\theta'}(\hat{\theta})(\tilde{\theta} - \hat{\theta}) + \mathcal{O}_p(n^{-3/2}).$$

Since $\dot{\ell}_{\theta}(\hat{\theta}) = \mathbf{0}$, upon rearranging:

$$\begin{aligned} 2\{\ell(\hat{\theta}) - \ell(\tilde{\theta})\} &= (\hat{\theta} - \tilde{\theta})' \{ -\ddot{\ell}_{\theta\theta'}(\hat{\theta}) \} (\hat{\theta} - \tilde{\theta}) + \mathcal{O}_p(n^{-3/2}) \\ &= n(\hat{\theta} - \tilde{\theta})' \{ -n^{-1} \ddot{\ell}_{\theta\theta'}(\hat{\theta}) \} (\hat{\theta} - \tilde{\theta}) + \mathcal{O}_p(n^{-3/2}) \\ &= n(\hat{\theta} - \tilde{\theta})' \mathbf{A}_0 (\hat{\theta} - \tilde{\theta}) + o_p(1). \end{aligned}$$

Taylor expand the score at the constrained MLE to obtain:

$$\dot{\ell}_{\theta}(\tilde{\theta}) = \dot{\ell}_{\theta}(\hat{\theta}) + \ddot{\ell}_{\theta\theta'}(\hat{\theta})(\tilde{\theta} - \hat{\theta}) + \mathcal{O}_p(n^{-1}).$$

Since $\dot{\ell}_{\theta}(\hat{\theta}) = \mathbf{0}$, upon rearranging:

$$\begin{aligned} \sqrt{n}(\hat{\theta} - \tilde{\theta}) &= \{ -n^{-1} \ddot{\ell}_{\theta\theta'}(\hat{\theta}) \}^{-1} \frac{1}{\sqrt{n}} \dot{\ell}_{\theta}(\tilde{\theta}) + \mathcal{O}_p(n^{-1}) \\ &= \mathbf{A}_0^{-1} \frac{1}{\sqrt{n}} \dot{\ell}_{\theta}(\tilde{\theta}) + o_p(1) \\ &= \mathbf{A}_0^{-1} \frac{1}{\sqrt{n}} \begin{pmatrix} \dot{\ell}_{\beta}(\beta_0, \tilde{\alpha}) \\ \dot{\ell}_{\alpha}(\beta_0, \tilde{\alpha}) \end{pmatrix} + o_p(1) \\ &= \mathbf{A}_0^{-1} \frac{1}{\sqrt{n}} \begin{pmatrix} \dot{\ell}_{\beta}(\beta_0, \tilde{\alpha}) \\ \mathbf{0} \end{pmatrix} + o_p(1). \end{aligned}$$

Recall from the score statistic that:

$$\frac{1}{\sqrt{n}} \dot{\ell}_{\beta}(\beta_0, \tilde{\alpha}) \xrightarrow{\mathcal{L}} N(\mathbf{0}, \mathbf{C}_0 \mathbf{B}_0 \mathbf{C}_0').$$

Thus:

$$\sqrt{n}(\hat{\theta} - \tilde{\theta}) \xrightarrow{\mathcal{L}} \mathbf{A}_0^{-1} \begin{pmatrix} N(\mathbf{0}, \mathbf{C}_0 \mathbf{B}_0 \mathbf{C}_0') \\ \mathbf{0} \end{pmatrix}.$$

The limiting distribution of the quadratic form is:

$$\begin{aligned} 2\{\ell(\hat{\boldsymbol{\theta}}) - \ell(\tilde{\boldsymbol{\theta}})\} &= n(\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}})' \mathbf{L}\mathbf{L}'(\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}}) + o_p(1), \\ &= \{\sqrt{n}(\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}})\}' \mathbf{A}_0 \{\sqrt{n}(\hat{\boldsymbol{\theta}} - \tilde{\boldsymbol{\theta}})\} + o_p(1) \\ &\xrightarrow{\mathcal{L}} (\boldsymbol{\omega}, \mathbf{0})' \mathbf{A}_0^{-1} \mathbf{A}_0 \mathbf{A}_0^{-1} \begin{pmatrix} \boldsymbol{\omega} \\ \mathbf{0} \end{pmatrix} = \boldsymbol{\omega}' (\mathbf{A}_0^{-1})_{\beta\beta'} \boldsymbol{\omega}, \end{aligned}$$

where $\boldsymbol{\omega} \sim N(\mathbf{0}, \mathbf{C}_0 \mathbf{B}_0 \mathbf{C}_0')$ and $(\mathbf{A}_0^{-1})_{\beta\beta'}$ is the $p \times p$ sub-matrix of $\mathbf{A}_0^{-1}$ corresponding to β . Let $\mathbf{L}\mathbf{L}'$ denote the Cholesky decomposition of $(\mathbf{A}_0^{-1})_{\beta\beta'}$. Then:

$$\boldsymbol{\omega}' (\mathbf{A}_0^{-1})_{\beta\beta'} \boldsymbol{\omega} = \boldsymbol{\omega}' \mathbf{L}\mathbf{L}' \boldsymbol{\omega} \stackrel{d}{=} \sum_{j=1}^p \lambda_j \chi_1^2,$$

where $(\lambda_1, \dots, \lambda_p)$ are the eigenvalues of $\mathbf{L}' \mathbf{C}_0 \mathbf{B}_0 \mathbf{C}_0' \mathbf{L}$. ■

Corollary 4.2. Within the exponential family $\mathbf{C}_0 \mathbf{B}_0 \mathbf{C}_0' = (\mathbf{A}_0^{-1})_{\beta\beta'}^{-1}$, such that:

$$(\mathbf{A}_0^{-1})_{\beta\beta'}^{1/2} (\mathbf{A}_0^{-1})_{\beta\beta'}^{-1} (\mathbf{A}_0^{-1})_{\beta\beta'}^{1/2} = \mathbf{I}.$$

Consequently, $\lambda_j = 1$ for each j , and:

$$2\{\ell(\hat{\boldsymbol{\theta}}) - \ell(\tilde{\boldsymbol{\theta}})\} \xrightarrow{\mathcal{L}} \chi_p^2. \quad \clubsuit$$